

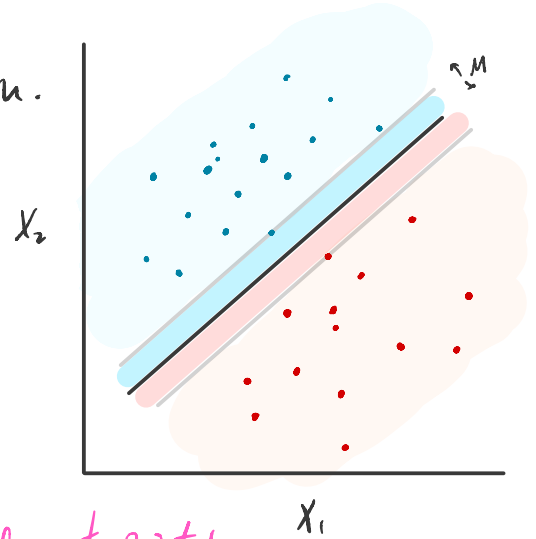
SUPPORT VECTOR MACHINES

Supervised technique used for classification.

- Hard Margin:

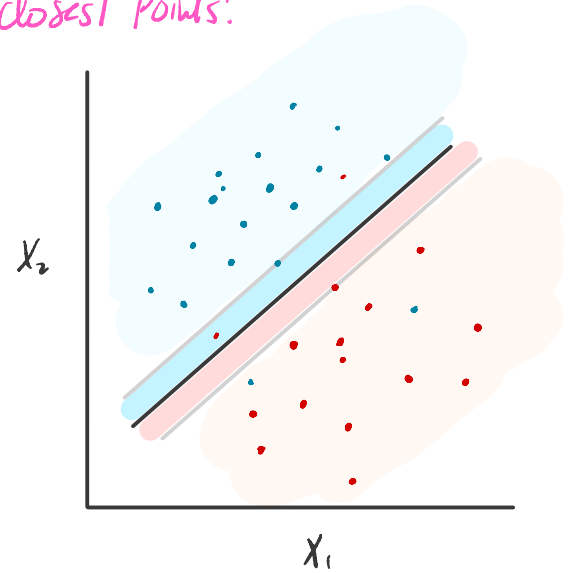
① Insert a separating hyperplane between the two classes

② maximize the margin M !



* notice the boundary only depends on the closest points!

so if we add a new point near the boundary the whole boundary will change!



- Soft Margin:

allow for some violations to preserve the accuracy of most data points

- what if data is not linearly separable? The kernel trick!

